MEEG304 User Manual

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OVERVIEW

The pill bottle filling system is a compact three-station machine built inside an 80-20 aluminum frame and designed so that two non-technical operators can set up and begin operation in under five minutes. The machine is built vertically for ease of use, storage capabilities, and a minimal footprint. At the top of the machine sits a rotary descrambler, which accepts up to 16 bottles (per batch), and through centripetal motion, guides one bottle at a time into a horizontal chute where a break beam sensor controls how many bottles are in the chute to prevent jamming. The horizontal chute features an end cap that ensures the bottles maintain the correct orientation throughout the rest of the system. A linear actuator extends through the horizontal chute to push the bottles into the vertical chute (again, in the correct orientation) into a secondary rotary table. This rotary table takes the bottles to a weighing station where their weight is zeroed. Simultaneously, the pill filler substation descrambles pills in a funnel using a motorized agitator and uses a servo to move two plates. When the plates are moved, pills fall from the funnel into a chamber with three casings that are the height of exactly ten pills each. This ensures the bottle receives thirty pills, no more and no less, for consumer safety and company liability. Once this operation has been completed, the plates close again, and the pills are dropped into the zeroed bottle using gravity. The load cell then verifies that the correct number of pills has entered the bottle. Should the pill filling operation provide more or less than thirty pills, the load cell will send this data to the rejector station. This communication is crucial for ensuring the machine does not accept misfilled bottles. Once at the rejector station, depending on the load cell measurement, a turnstile connected to a high-strength servo will either rotate, kicking the bottle into a bin for further inspection, or accept the bottle by doing nothing. If the bottle is accepted, it continues through the rotary table until it falls a short distance onto a conveyor belt and is sent to the next station to be capped and tightened. Daily use is made simple with this design, machined for automation, and with little to no operator interference. The machine features an easy-to-access emergency stop button. Simply press the button to kill power to the system and shut it down. Rotate the button to the right to reset this operation. A green button is located by the machine's electronics for technical ease. Once you hit start, microcontrollers coordinate all of the motion within the machine until you press the stop button. This manual walks the user through assembly steps, routine maintenance, and a troubleshooting guide to cover all commonly experienced problems, including but not limited to jamming or pill miscounts. A list will show readily available replacement parts for the linear actuator, stepper motors, sensors, load cell, and other electronics that may need to be replaced. This list ensures that you will have minimal downtime and won't get behind on batches, so your customers can be happy. Together, these features deliver a plug-and-play workspace solution that can be scaled or serviced with off-the-shelf components, giving users a reliable path from bulk product to sealed, measured bottles in just minutes. Figure 1 shows the full system assembly with the major components labeled.

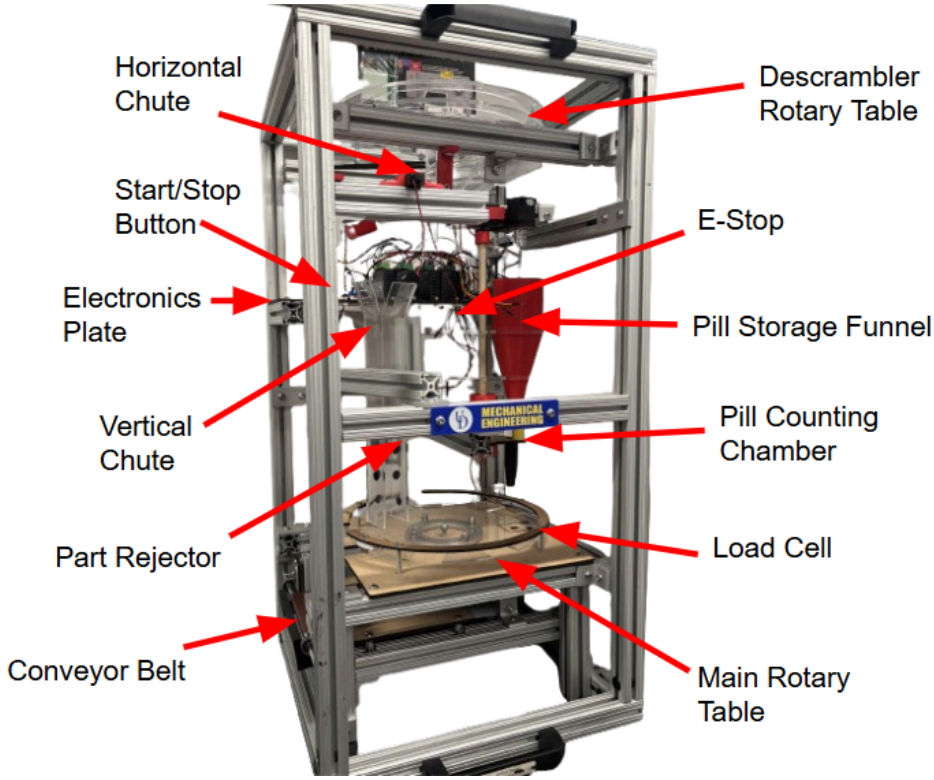

Figure 1: Full system

ASSEMBLY AND DISASSEMBLY

Bottle Descrambler:

For the bottle descrambler subsystem assembly, the linear actuator, top rotary table zip ties, and shaft set screws need to be assembled. Use a 3/16 Allen key to loosen the two bolts shown in Figure 2, securing the linear actuator to its respective piece of 80-20. Slide the linear actuator towards the frame until the very end of the actuator is no longer inside the horizontal chute. Tighten bolts to secure the linear actuator.

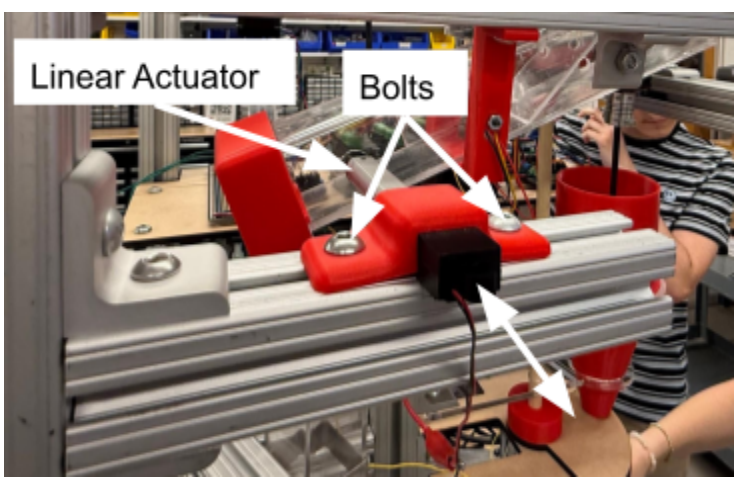

Figure 2: Linear Actuator

Ensure that the zip ties attached to the top rotary table are secure and that the ends are parallel with the surface of the rotary table. Also, ensure that the set screws on the shaft coupler and gear are tightened. For disassembly, loosen the two bolts securing the linear actuator and slide it forward until the back of the linear actuator is fully inside the 80/20 frame. Tighten the bolts to secure the linear actuator for moving/storage.

Conveyor Belt:
Connect the conveyor ledge in between the two vertical front 80-20 with two bolts. Flip up when the system is in motion to act as a gap filler when bottles travel to the next station (Figure 3).

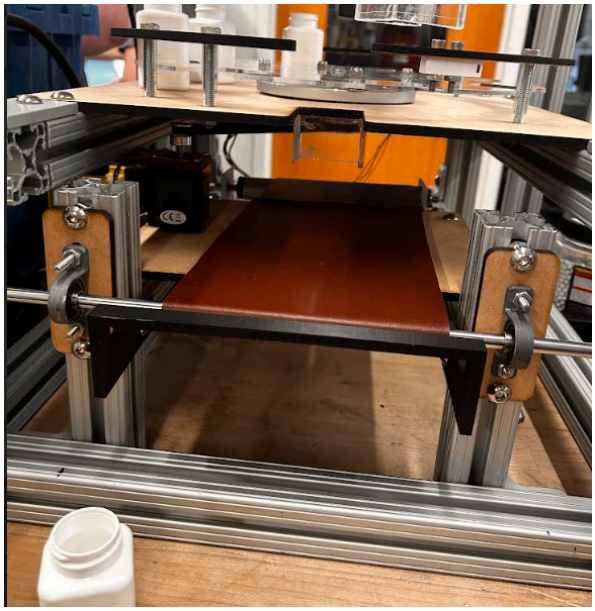

Figure 3: Front conveyor set up with ledge

OPERATING INSTRUCTIONS
Follow the steps below to set up and operate the machine safely.
1. Load the Bottles
Place up to 16 bottles into the descrambler rotary table at the top of the machine as shown in Figure 4. The table and bottles do not need to be precisely aligned; simply ensure they don't spill out of the table. It is okay if bottles fall into the horizontal chute when initially loaded.

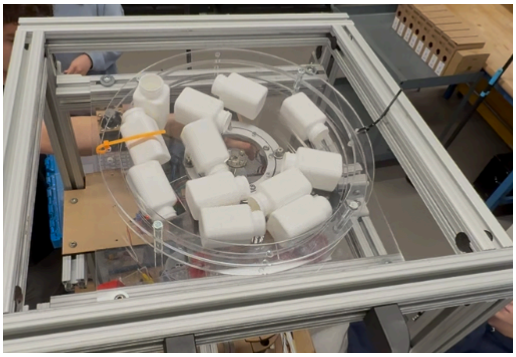

Figure 4: Descrambler Table loaded with bottles

Use one bottle to align the main rotary table at the bottom of the machine. With the bottle flush against the front of any slot in the main rotary table, move it to the starting line labeled “bottle.” Do not move the main rotary table between this step and starting the machine, or it will need to be realigned.

2. Fill the Pill Funnel

Pour approximately 500 pills into the funnel at the top of the system. Do not overfill, as this may affect the system’s ability to descramble and properly count the pills.

3. Power and Load Cell Calibration

Ensure the power cord is plugged into a standard wall outlet. Release the Emergency Stop by twisting the red button to allow electrical power to reach the system. Connect to the Arduino Mega and upload the system code, then connect to the Arduino Uno and upload the code necessary for calibration. Ensure that nothing interacts with the load cell, and confirm calibration by looking at the ADC value read out, confirming that the value is under 100 (it can be negative).

4. Prepare for Operation

Ensure that the rejection station is next to the rejection bin to prevent spillage of pills from rejected bottles. The machine must be aligned with the back station capping machine. To do this, ensure that the conveyor belt ledge is level with the back station inlet conveyor.

5. Start the Machine

Once all components are prepared and calibration is confirmed, press the green start button to begin operation. The system will automatically process the bottles and dispense pills as programmed. To stop the system, press the Emergency Stop Button.

TROUBLESHOOTING GUIDE

Descrambler:

- **Bottles may get stuck in the middle of the descrambler rotary table.** They can be manually moved from the center of the table so that they fall into the table slots.
- **Bottles may get stuck in the walls as they fall into the horizontal chute.** To help them fall into the chute, the bottle can be pushed from the back of the chute. If this does not unjam the bottle, turn off the machine's power. Then rotate the table so one of the openings is above the horizontal chute, and pull or push on the bottle to unjam it. Ensure the main rotary table is still aligned before restarting the machine.
- **Bottles may get stuck between the horizontal and vertical chute.** If the bottleneck is face down and does not slide all the way into the slot, push the bottle into the tube so that it slides all the way down and can be expelled by the linear actuator. If a bottle gets stuck in the funnel at the top of the vertical chute, it may unjam itself with no intervention or when the next bottle falls onto it. If it does not unjam itself, manually move the bottle so that it falls into the vertical chute.
- **A bottle may get stuck inside the vertical chute.** It may unjam itself when the next bottle falls on top of it or with the natural vibrations of the system running. If the bottle

remains jammed, tapping on the sides of the tube can often unjam it. Otherwise, drop another bottle on top to push the stuck bottle down.

- **Bottles may tip over as they exit the bottom of the vertical chute.** This is due to the zip tie fatiguing or breaking. Remove the existing zip tie. Attach a new zip tie and zip tie mount. Cut the zip tie to the same length as the removed zip tie.

Pill Filler:

- **Pills may jam plates, causing them to get stuck.** Manually shaking the plates slightly can fix the pill positioning, ensuring a smooth rotation for the next bottle.
- **Pills may fly out of the pill pocket.** If this is happening, decrease the speed of the pill descrambler's rotation. If it rotates for too long or too fast, it will try to push too many pills down, causing jamming between the top plate and the stationary.
- **The pill filler might fill fewer than 30 pills.** This could be caused by a misalignment between the bottom of the funnel and the top of the counting chamber. Check this and make adjustments as needed. It could also be caused by the pill descrambler not rotating for long enough.

Bottle Rejection:

- **The load cell may experience electrical or mechanical noise, causing a poor measurement.** Ensure that all electrical connections are secure and that the system is stable while in use.
- **The load cell may experience calibration drift, leading to inaccurate measurements.** Before starting to use the machine, the initial load cell calibration ADC constant should be as minimal as possible.
- **The servo may experience failure with extended use.** To prevent this, ensure that any bottle rejected has as little resistance as possible. This can be done by ensuring the system's alignment is correct. Otherwise, the servo will be put under strain while attempting to reject a bottle through the solid guard rail.

Electronics:

- **The pill servo no longer moves on bottle detection.** If this occurs, check the connections to the load cell and ensure that the Arduino UNO is connected to the computer properly. Lack of signal to the Arduino UNO can cause issues with the communication between the load cell and the system code.

REPLACEMENT PARTS

Replacement Part	Link
Linear Actuator	Link
Zip Ties	Link
Stepper Motors	Link
Break Beam Sensor	Link
Servo Motors	Link